Transfer: waves and signals

antocciphysics | TEACHER KEY | CPA

Learning target: Choose appropriate wave models and explain what a signal establishes.

Suggested first pass: 1, 2, 3, 7. Add independent, mixed and stretch work from the bank as time and student evidence permit. Estimated times are planning estimates, not completion requirements.

01. Bring back an earlier idea (retrieve) *Review: Unit 6, Day 15; 3 course days later.*
2500 m; range is half the full path.

02. Explain a worked example (worked)
Round-trip timing plus propagation speed gives range; a separate frequency comparison helps determine radial motion.

03. Look and explain (model)
Amplitude: size from equilibrium; wavelength: distance between matching phase points.

04. Build the solution (practice)
10 m/s; range 20 m.

05. Use the idea (reason)
No; amplitude/intensity change is distinct from frequency.

06. Choose a method (mixed) *Review: Unit 6, Day 6; 12 course days later.*
The 6 cm wave generally spreads more because wavelength is larger relative to the same opening.

07. Exit check (exit)
Toward normal; frequency unchanged at stationary boundary.

08. STRETCH: Apply it another way (stretch)
Sound reflector 85 m; light reflector 300 m. Shorter time need not mean shorter range when signal speeds differ.

Teaching note: Self-contained daily transfer worksheet. Keep any separately assigned formal assessment independent.

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01. Bring back an earlier idea (retrieve) *Review: Unit 6, Day 15; 3 course days later.*
At the same wave speed, equal range gives equal out-and-back time. Doppler shift depends on radial velocity, so motion can differ.

02. Explain a worked example (worked)
Round-trip timing plus propagation speed gives range; a separate frequency comparison helps determine radial motion.

03. Represent and interpret (model)
Amplitude: size from equilibrium; wavelength: distance between matching phase points.

04. Solve and justify (practice)
1.8×10^{6} m;λ=.10 m.

05. Analyze the reasoning (reason)
No; amplitude/intensity change is distinct from frequency.

06. Choose a method (mixed) *Review: Unit 6, Day 6; 12 course days later.*
Wavelength becomes 1/4; typically less spreading when the opening is large relative to the new wavelength.

07. Exit check (exit)
Thermal peak halves by Wien; soundλhalves at fixed sound speed.

08. STRETCH: Apply it another way (stretch)
1500 and 1440 m; radial velocity -300 m/s. Electromagnetic propagation remains about 3×10^{8} m/s in air under the model.

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